

Functional Requirements Specification (FRS)

System: Jima — CARAVAN Lounge Restaurant Management System
Document: FRS v1.1 · **Updated:** August 20, 2026
Related documents: SRS (02), SDD (03), User Manual (04), Handoff (05), deployment guide (06), and operational runbooks (07–12)

1. Purpose

This document defines the functional requirements of the Jima Restaurant Management System — a full-stack web application that manages the daily operations of CARAVAN Lounge: ordering, kitchen workflow, payments, inventory, staff, branches, and reporting.

Verified implementation baseline: the registered Jima web artifact runs the real Next.js 15 / React 19 frontend in `artifacts/nextjs-app`. Browser requests use `/backend/api/*`, which Next.js rewrites to the NestJS service's `/api/*`. NestJS and TypeORM own all PostgreSQL access. The separate `artifacts/api-server` is not the restaurant application API.

Requirements versus evidence: the items below define required behavior. A requirement must not be treated as production-verified unless it appears as passed evidence in document 12.

2. User Roles

Role	Description
Admin	Full system access; manages all data, users, and configuration.

Owner	Business owner; same operational visibility as manager, including analytics dashboard.
Manager	Runs daily operations; manages menu, staff, tables, branches, inventory, and reports.
Coordinator	Confirms incoming orders and assigns them to chefs; can cancel orders.
Waiter	Takes orders at tables (POS), adds items to open orders, tracks order progress.
Chef	Works the kitchen board: accepts, prepares, and marks orders ready.
Cashier	Collects payments, closes orders, views sales and items-purchased reports.
Storekeeper	Manages stock levels, receives goods, handles item requests and adjustments.

3. Functional Requirements

3.1 Authentication & Authorization

ID	Requirement
FR-A01	Users log in with e-mail and password; the system returns a JWT access token.
FR-A02	Every protected API call requires a valid <code>Authorization: Bearer</code> token. Login and health endpoints are intentional public exceptions; administrative seed operations must remain protected.
FR-A03	Menu entries and pages are restricted per role (role-based access control).
FR-A04	Passwords are stored hashed (bcrypt), never in plain text.

3.2 Orders & POS

ID	Requirement
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FR-O01	Waiters create orders through a POS screen: pick a table (or take-away), select menu items by category, set quantities and notes.
FR-O02	Orders follow the lifecycle: pending → confirmed → preparing → ready → served → paid , with cancelled possible before "ready".
FR-O03	The coordinator confirms a pending order and assigns a chef to it.
FR-O04	Waiters may append items to any order that is not yet paid or cancelled.
FR-O05	The orders list shows table, chef, item count, time, total, payment method, and status; rows open a detail view.
FR-O06	Orders can be filtered by status, waiter, table, item, date, time, and payment method.
FR-O07	The list and open detail view refresh automatically every 10 seconds.

3.3 Kitchen

ID	Requirement
FR-K01	The kitchen board shows confirmed orders queued per chef, with tabs per chef.
FR-K02	Chefs move orders through accept → preparing → ready.
FR-K03	The Order Progress Board displays four live columns: Requested, Received & Confirmed, In Progress, Completed (ready only).
FR-K04	Orders waiting more than 20 minutes are visually flagged.

3.4 Payments

ID	Requirement
FR-P01	Cashiers record payments against served orders with method cash, card, or wallet, including change given.
FR-P02	Paying an order marks it paid and frees its table.
FR-P03	

	Daily reports summarize revenue, transaction counts, and per-method breakdowns.
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3.5 Menu Management

ID	Requirement
FR-M01	Managers maintain menu categories and items (name, price, description, availability, preparation time).
FR-M02	Unavailable items are hidden from the POS.

3.6 Inventory & Store

ID	Requirement
FR-I01	Inventory items track unit, current stock, minimum stock, reorder point, unit cost, category, and expiry date.
FR-I02	Low-stock alerts flag items below minimum (critical when at or below half the minimum, or zero).
FR-I03	Suppliers and purchase orders are managed with status tracking.
FR-I04	Stock adjustments record additions, deductions, and waste with reasons and the acting user.
FR-I05	Staff submit item requests; managers/storekeepers approve, issue, and track them, with low-stock / expiry badges shown on request rows.

3.7 Staff, Branches & Tables

ID	Requirement
FR-S01	Managers create and manage staff accounts with roles and branch assignment.
FR-S02	Restaurants and branches are managed hierarchically (restaurant → branches).
FR-S03	

	Tables have number, capacity, section, and live status (available / occupied).
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3.8 Dashboards & Reports

ID	Requirement
FR-D01	Admin/Owner/Manager see a combined dashboard: today's KPIs (sales, orders, average order value, in-progress, low stock, stock value), sales trend, top sellers, profitability, and kitchen performance — on a single page. KPI cards open detail popups.
FR-D02	Reports offer per-area export tabs (Order Board, Menu, Inventory, Item Requested, Staff, Branches) with filters and Excel / PDF export.
FR-D03	Cashiers see a reduced report view: Sales and Items Purchased only, filterable by type and date, exportable to Excel / PDF.
FR-D04	Role-specific landing: waiters, coordinators, and cashiers land on Orders/POS instead of the dashboard.

3.9 Notifications

ID	Requirement
FR-N01	The system generates notifications targeted at roles or specific users (e.g. new order, order ready) with read tracking.

4. Business Rules

- An order's total is the sum of its items (quantity × unit price) plus any service charge percentage.
- Only pending orders can be confirmed; only served orders can be paid.
- Cancelling is allowed only while an order is pending, confirmed, or preparing.
- Data is scoped by branch where applicable (orders, tables, staff, stock).

5. Verification Status

Functional development is active and major workflows have been exercised manually. Complete automated regression evidence for every role, branch boundary, order transition, payment, inventory operation, notification, report, and export is not yet recorded. Production release remains blocked until the verification matrix in document 12 is complete.